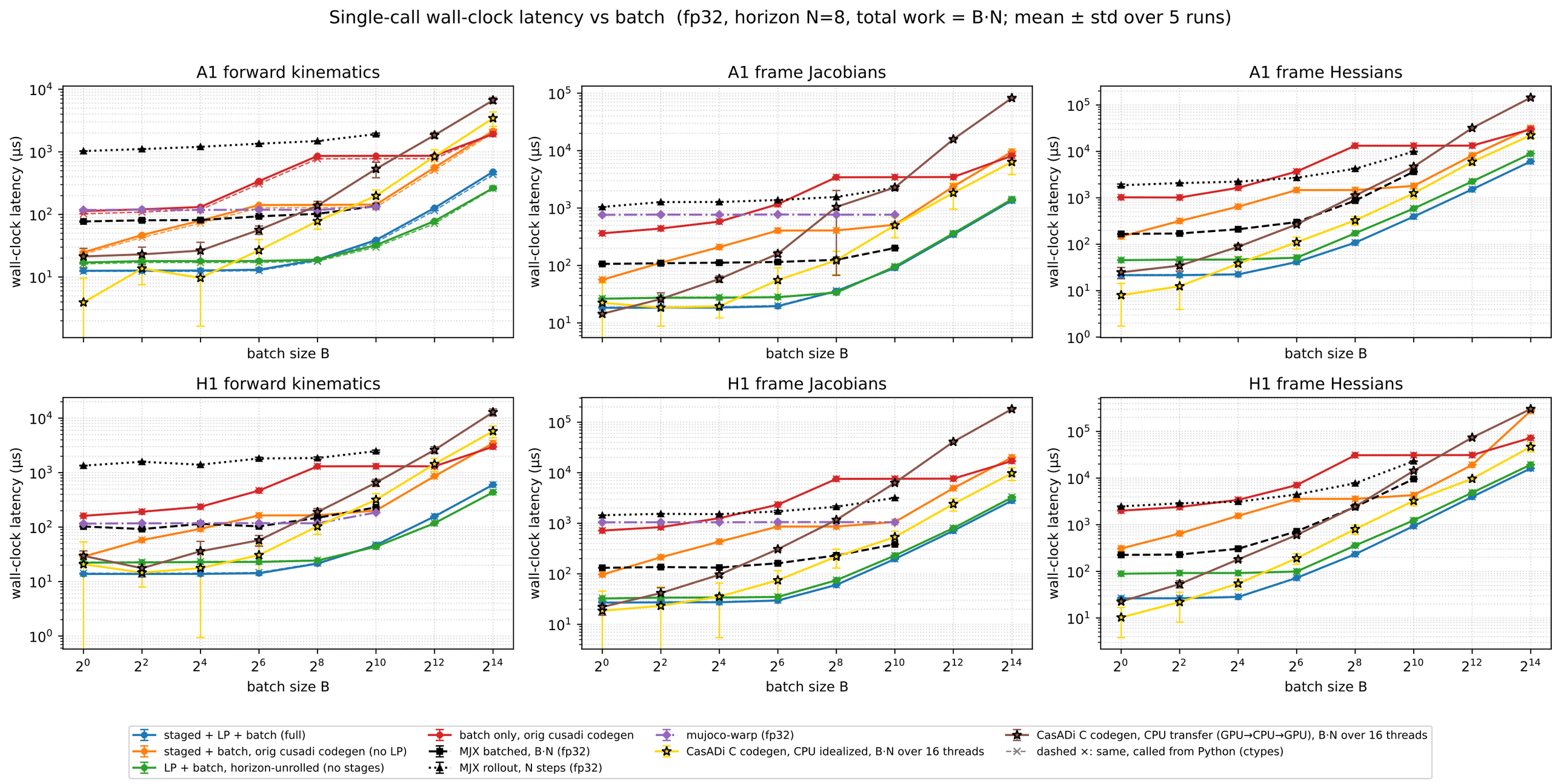

Single-call wall-clock latency vs batch (fp32, horizon N=8, total work = B·N; mean ± std over 5 runs)
A1 forward kinematics
A1 frame Jacobians
A1 frame Hessians
H1 forward kinematics
H1 frame Jacobians
H1 frame Hessians
wall-clock latency (µs)
batch size B
staged + LP + batch (full)
staged + batch, orig cusadi codegen (no LP)
LP + batch, horizon-unrolled (no stages)
batch only, orig cusadi codegen
MJX batched, B·N (fp32)
MJX rollout, N steps (fp32)
mujoco-warp (fp32)
CasADi C codegen, CPU idealized, B·N over 16 threads
CasADi C codegen, CPU transfer (GPU→CPU→GPU), B·N over 16 threads
dashed ×: same, called from Python (ctypes)